

Atorvastatin inhibition of the mevalonate pathway shows promise as a novel anti-cancer therapeutic in in vivo models of hormone driven cancers

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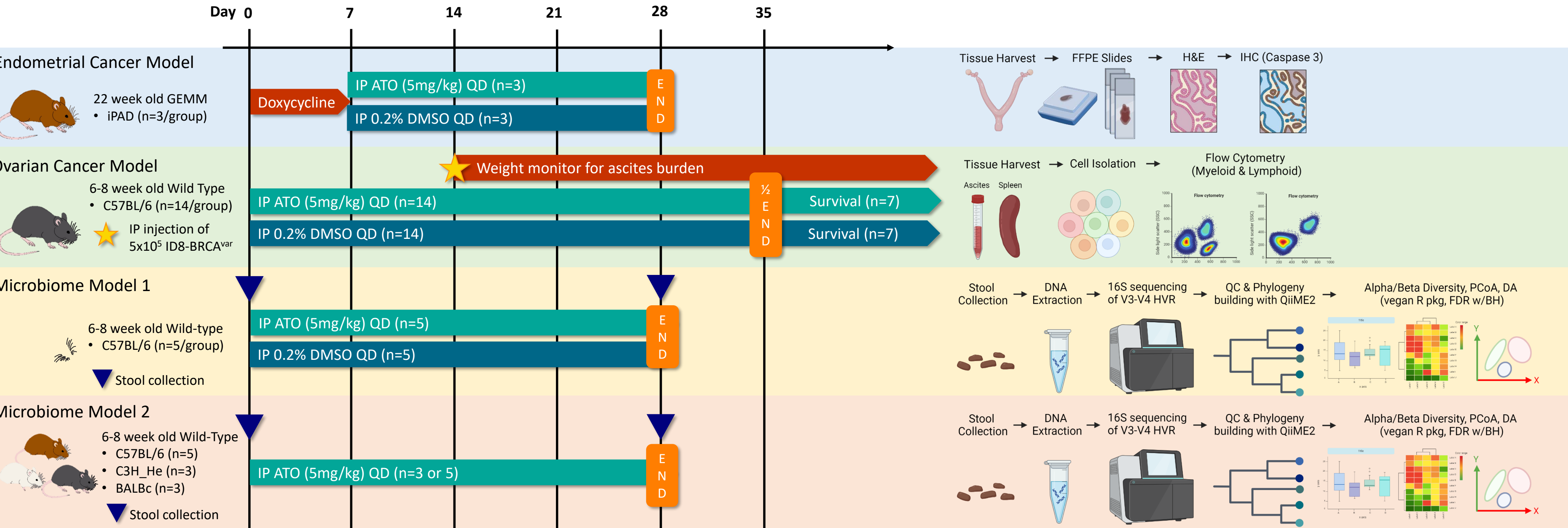


Abstract

Female gynecological cancers remain a challenge for women in the developed world today. Over the last few decades, the incidence and mortality of endometrial cancer (EC) has been steadily rising, with diagnosis of disease increasing worldwide. Moreover, there has been only marginal improvements in mortality for ovarian cancer (OC), despite recent advancements in diagnostic technologies and novel treatments. This presents a tremendous public health concern for women worldwide and research is urgently needed to develop preventative strategies for women to reduce the risk and mortality of reproductive cancers. Repurposing of existing, non-cancer-related drugs for use as anti-cancer agents offers an *avant-garde* approach to identifying new prevention strategies for gynecological cancers. Statins, a mevalonate pathway antagonist, are widely used to treat and prevent hypercholesterolemia, however, recent evidence has highlighted statins exhibit pleiotropic anti-tumor effects. Furthermore, statins have been shown to induce large scale alterations of intestinal enterotypes. This is uniquely important as recent evidence has shown gut health significantly impacts cancer outcomes. Here, we present data from statin treated OC and EC murine models, where we observe statin treatment, in the absence of standard of care, is able to modestly improve gynecological tumor outcomes. Furthermore, we characterize how statin treatment modulates the intestinal microbial landscape toward a more homeostatic and diverse microbiome. In summary, we show that statin treatment has potential to serve as a novel anti-cancer therapy for hormonally driven cancers, and that favorable modulation of the intestinal microbiome may be one mechanism by which statins are able to provide benefit.

Methods

Endometrial and ovarian cancer murine models were dosed with Atorvastatin (5mg/kg) or DMSO (0.2%) and monitored for tumor progression. Tumor and tissues were collected at designated endpoints and further analyzed via various histopathological methods, including cancer staging of H&E slides, IHC, and immune profiling via flow cytometry. To investigate how statin treatment affected the microbiome, wild-type mice of various strains were dosed with ATO (5mg/kg) or DMSO (0.2%) for 4 weeks. Baseline and endpoint stool samples were collected, and 16S sequencing was done for genus level identification of intestinal taxa.



Statin favorably modifies the intestinal flora to a more homeostatic state

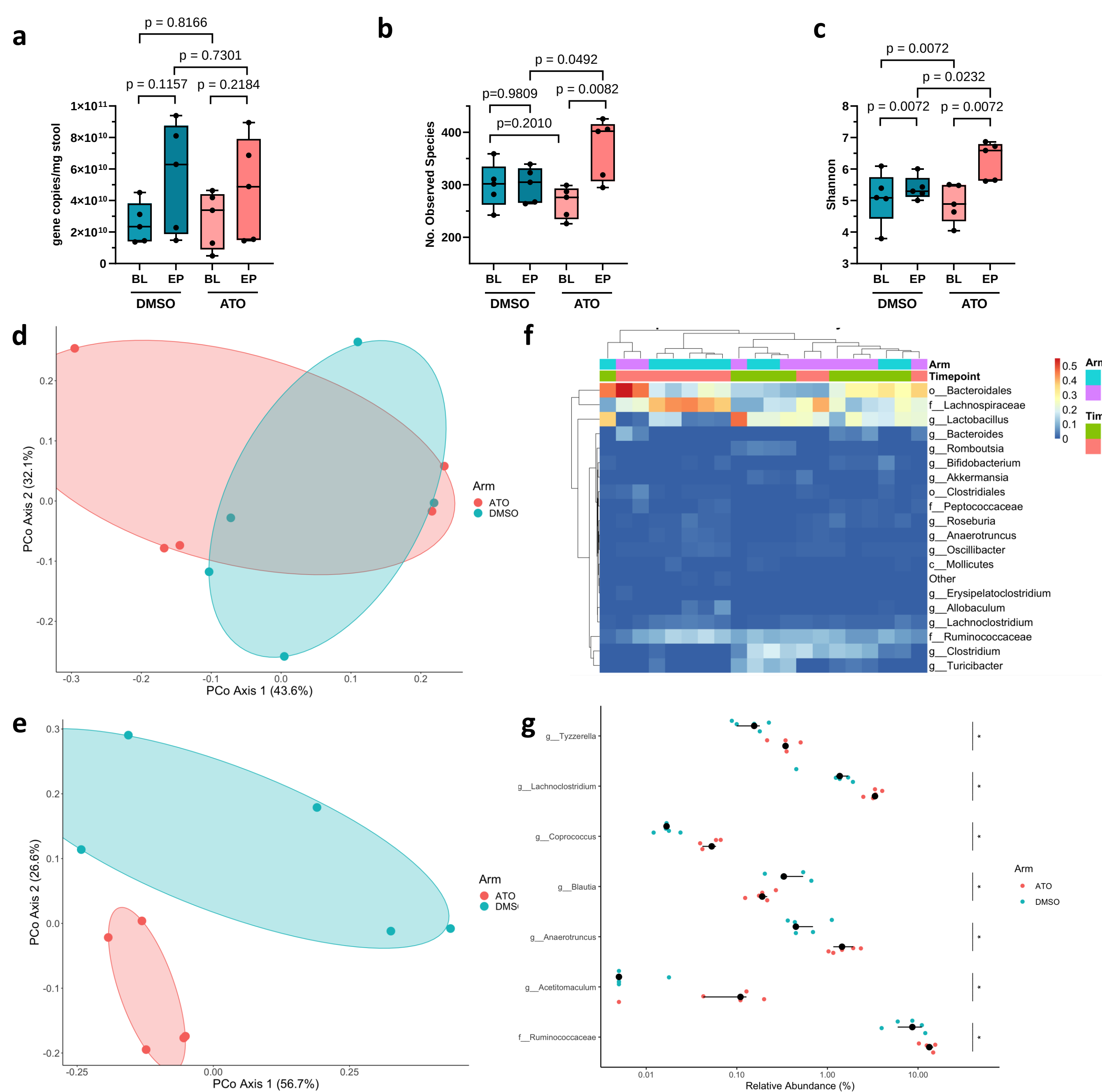


Fig. 4. ATO treatment favorably shifts the intestinal microbiome. (a) Average Absolute Abundance (b-c) Average alpha diversity metrics. (d-e) PCoA plots generated with Bray-Curtis Dissimilarity at Baseline (d) and Endpoint (e). (f) Heatmap of 20 most abundant taxa. Samples clustered by Bray-Curtis dissimilarity. (g) Scatter plot of taxa that are significantly different in abundance. All data are Min to Max. * if p<0.05, ** if p<0.01, *** if p<0.001, 50%CI

Genetic background has minimal effect on statin's ability to modulate the microbiome

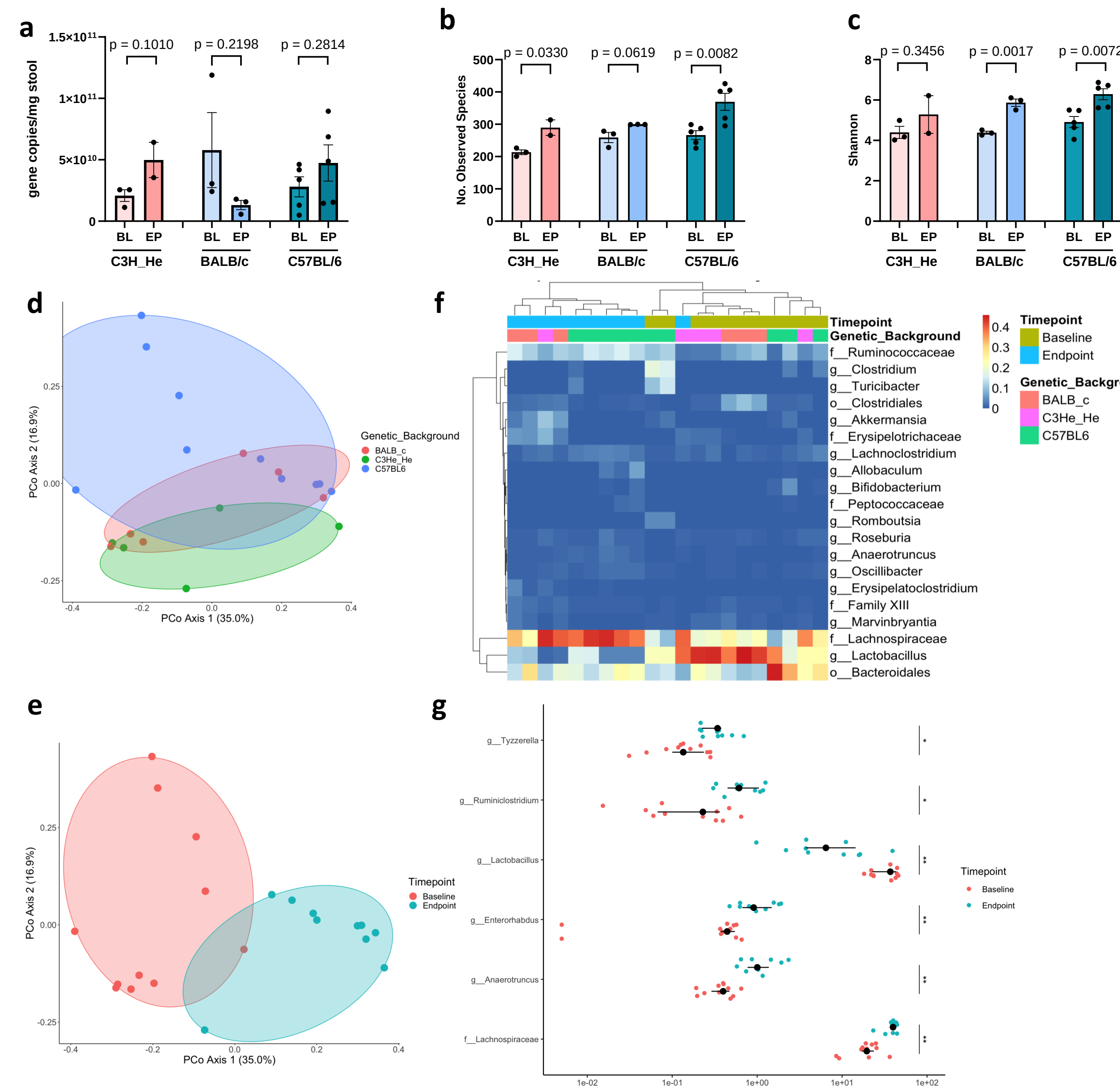


Fig. 5. ATO treatment shifts the microbiome similarly, regardless of genetic background. (a) Average Absolute Abundance (b-c) Average alpha diversity metrics. (d-e) PCoA plots generated with Bray-Curtis Dissimilarity. Samples colored by mouse strain (d) and timepoint (e). (f) Heatmap of 20 most abundant taxa. Samples clustered by Bray-Curtis dissimilarity. (g) Scatter plot of taxa that are significantly different in abundance. All data are Min to Max. * if p<0.05, ** if p<0.01, *** if p<0.001, 50%CI

Results

Atorvastatin treatment reduced metastasis and increased apoptosis in an inducible model of endometrial cancer

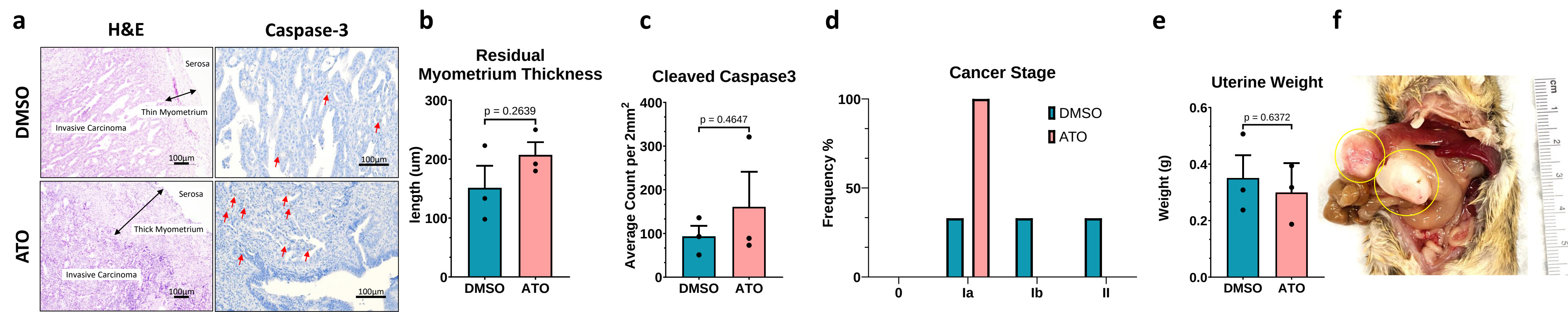


Fig. 1. ATO treatment is associated with reduced cancer stage and increased apoptosis in an EC model. (a) Representative H&E and Caspase 3 IHC. Black arrows show residual myometrium thickness. Red arrows point to Caspase 3 positive loci. (b) Average residual myometrium thickness. (c) Average Caspase 3 positive loci per 2mm² tissue area. (d) Bar graph representing tumor stage. (e) Average uterine weight normalized to total mouse weight. (f) Gross image of metastatic tumors on intestine of DMSO treated mouse (yellow circles). All data are means±SEM. Scale bar 100µm.

Prophylactic statin administration improved tumor free survival but not tumor burden in an syngeneic murine model of ovarian cancer.

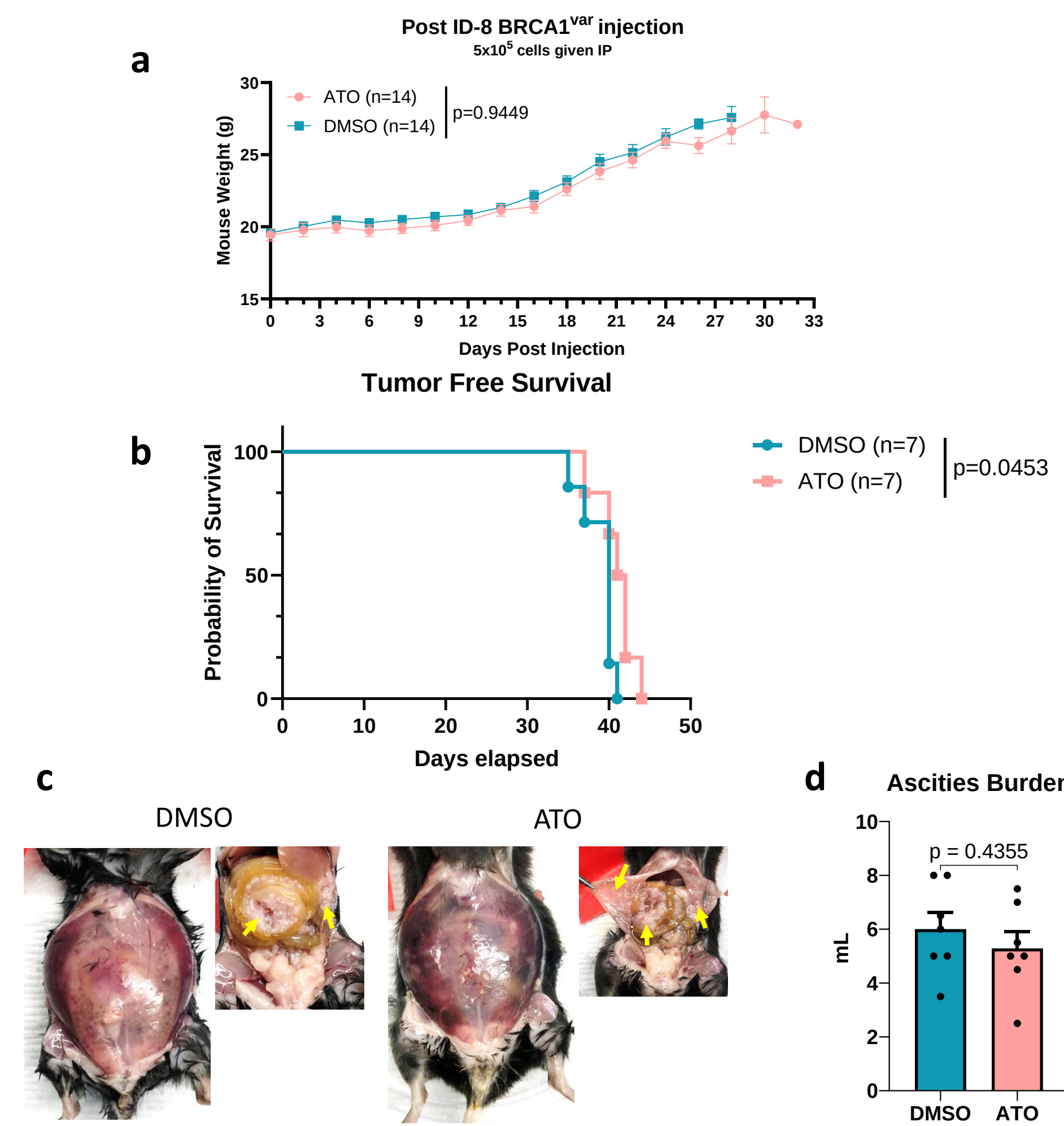


Fig. 2. ATO treatment did not improve tumor burden but did improve tumor-free survival in an OC model. (a) Growth curves post ID8-BRCA^{WT} injection. (b) Kaplan-Meier curves showing tumor free survival of ATO (pink) or DMSO (teal) treated mice. (c) Gross image of ascites fluid in peritoneal cavity and abundant peritoneal metastatic lesions (yellow arrows). (d) Average ascites tumor burden. All data are means±SEM.

Statin treatment increased splenic activated CD8+ T-cell populations, but had minimal effect on myeloid cell populations

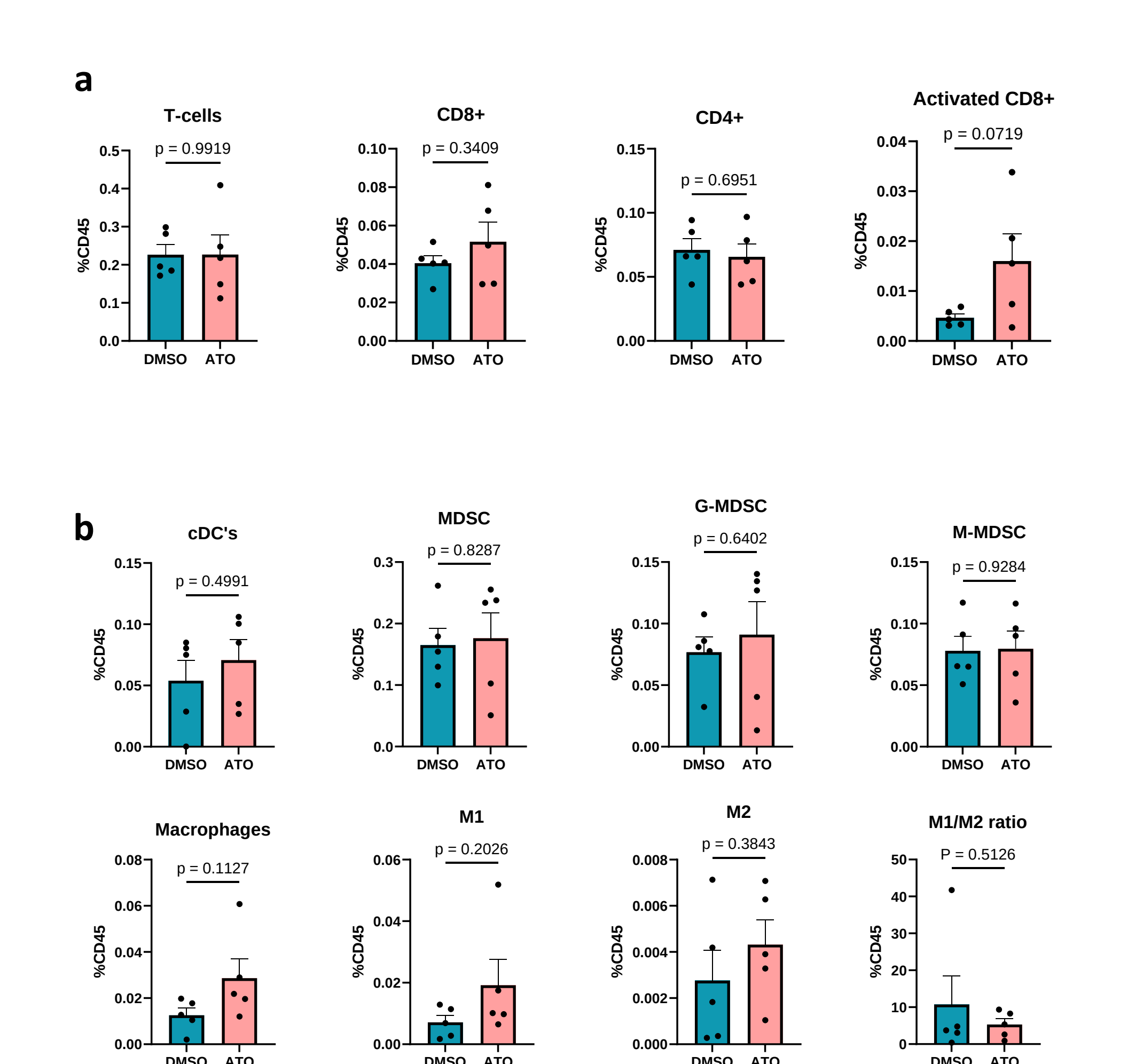


Fig. 3. ATO treatment increased splenic activated CTLs but not myeloid cells. (a) Bar charts of lymphoid cell populations as a percent of total CD45+ immune cells. (b) Bar charts of myeloid cell populations as a percent of total CD45+ immune cells. All data are means±SEM.

Summary and Conclusions

Here, we present data from statin treated ovarian and endometrial cancer murine models, where we observe statin treatment is associated with modest improvement in tumor outcomes. In the EC model, we saw statin treatment reduced tumor stage and increased apoptosis as measured by Cleaved Caspase3 staining. We also found that only our DMSO treated mice developed metastatic lesions. In our OC model, we found that statin exhibited mild chemoprotective effects with statin treated mice having significantly increased tumor free survival.

Furthermore, we describe how statin use alters the intestinal microbiome in commonly used mouse strains toward a more homeostatic state. In comparison to DMSO treated mice, we saw significantly increased abundance of many important bacteria associated with barrier protection, butyrate productions and gut health such as *Lachnospiraceae*, *Coprococcus*, *Acetivibacter*, and *Ruminococcaceae*. Lastly, we discovered that microbiome shifts were independent of mouse strain, indicating that statin's are able to modulate the microbiome similarly regardless of genetic background.

Overall, our results support further research to evaluate statin use as chemopreventive agent for gynecological cancers.

Acknowledgments

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